

Initialization

Set parameters $\theta = \{W^{(\ell)}, b^{(\ell)}\}_{\ell=1}^3$; sample collocation points $\{t_i\}_{i=1}^{N_c} \subset [0, 1]$

Forward Pass

Propagate both $\hat{y}(t_i)$ and $\hat{y}'(t_i)$ simultaneously through the network

Loss Computation

$$L = \underbrace{\frac{1}{N_c} \sum_{i=1}^{N_c} [\hat{y}'(t_i) + \hat{y}(t_i)]^2}_{L_R} + \lambda \underbrace{(\hat{y}(0) - 1)^2}_{L_{IC}}$$

Backpropagation

Compute $\nabla_{\theta} L$ via chain rule — *includes the product rule arising from the time derivative $\hat{y}'$*

Parameter Update

$$\theta \leftarrow \theta - \eta \nabla_{\theta} L$$

Repeat until convergence